10.7.1 Chikungunya Virus

The virus was first isolated from human patients of ***Aedes aegypti*** mosquitoes (Figure 10.9) from Tanzania in 1952. The name Chikungunya is derived from the native word for the disease in which the patient lies **'doubled up due to severe joint pains'**. The virus first appeared in India in 1963 in Calcutta, Madras and Other areas.

The disease presents as a sudden onset of fever, Crippling joint pains, lymphadenopathy and conjunctivitis. A **maculopapular rash** in common. The fever is typically biphasic with a period of remission after 1–6 days of fever. The vector is ***Aedes aegypti***. No animal reservoir has been identified. Antibody to the virus has been demonstrated in horses, cattle and other domestic animals.

Flavi viruses

The family flaviviridae contains only one genus **flavivirus**. They are smaller than alpha viruses, being 40nm in diameter. There are over 60 arthropod borne flava viruses classified as **mosquito-borne and tick borne viruses**. Examples of mosquito borne group known as encephalitis viruses they are St. Louis encephalitis Virus, Ilheus virus, west nile virus, murray valley encephalitis virus and Japanese encephalitis. Tick borne viruses ore classified in to tick brone encephalitis viruses and tick borne hemorrhagic fevers.



Figure 10.9: *Aedes aegypti*

10.7.2 Dengue

The name dengue is derived from the **'Swahili ki denga pepo'**, meaning a sudden **seizure by a demon**. Dengue fever is similar to the illness caused by chikungunya. Four types of dengue virus exist: **DEN1, DEN2, DEN3 and DEN4**.

HOTS

What is the best home remedy for dengue fever?

Dengue presents after an incubation period of 3-14 days as fever of sudden onset with headache, retrobulbarpain, conjunctival injection, pain in the back and limbs (break bone fever), lymphadenopathy and maculopapular rash. The fever is typically **biphasic** (saddle back) and lasts for 5–7 days. Dengue may be more serious forms with **hemorrhagic** manifestations (dengue Hemorrhagic fever) or with shock (dengue shock syndrome).

Dengue virus is transmitted from person to person by ***Aedes aegypti*** mosquitoes. The Incubation period is 8–10days. All four types of dengue virus are identified. Demonstration of circulating IgM antibody provides early diagnosis. IgM ELISA test offers reliable diagnosis. Difference between Dengue and Chikungunya is given in Table 10.2.

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